

BUILD-SUMMARY.md — what actually exists

Assembled 2026-08-05 from the repository at `main @ 545a2ac`, by reading the code and running it. Where a design document and the code disagree, **the code is reported and the disagreement is recorded in §5**. Nothing planned is described as existing.

How to read this. §1 is the map. §2 is the inventory with evidence and invocation. §3 is the configuration, verbatim from the files that hold it. §4 is the test estate. §5 is the gap list, and §5 is the part worth your time — an orientation document whose gap section is short is a document that did not look.

1. System map

Dependency direction is downward only. `port / gen / eval / artifacts` depend on `data`, which depends on `core`; nothing points back up. The product surface (`bundle`, `feed`, `serve`, `play`, `credibility`, `programme`) sits on top of `core` + `port` and is the only layer the browser sees.

```
Terrarium/
├─ src/ah/
│   ├── core/
│   │   ├── worldspect.py    the numeric core – no pandas, no network, no clock
│   │   ├── loader.py        pydantic mirror of schemas/worldspec-v1.2.schema.json
│   │   ├── validator.py     jsonschema-first then pydantic (a property test pins agreement)
│   │   ├── numericworld.py  V1-V12 coherence rules; V10/V11/V12 block → validate()
│   │   ├── engine.py        the narrative-free projection the engine consumes
│   │   ├── institution.py   the deterministic toy engine `toy-v0.3` → run_path/run_ensemble
│   │   ├── sleevestate.py   Step-0 toy institution (8 weights, no cash) → simulate_institution
│   │   ├── institutionstate.py Step-3 sleeve/vehicle state contract
│   │   └── digest.py        institution state for the twin
│   │   └── store/
│   │       ├── db.py        canonical JSON + SHA-256 over float64 tensors
│   │       ├── worlds.py    SQLite; append-only where it matters
│   │       ├── runrecords.py schema + triggers (chronicle UPDATE/DELETE refused)
│   │       ├── chronicle.py save_world refuses edits to engine-consumed fields
│   │       ├── sessions.py  RunRecords + verify_run (the anchor behind `ah replay`)
│   │       └── leaderboard.py append-only event log per world
│   │           submit_score/scores over db.py:92's
│   │           UNIQUE(world_id, seed, decision_alpha_version, participant)
│   ├── compiler/
│   │   ├── pipeline.py      scenario text → raw world document
│   │   ├── fixture_adapter.py validate → clamp → construct WorldSpec → process()
│   │   └── anthropic_adapter.py the offline compiler (50 committed fixtures)
│   │       the live compiler, imported only under --live
│   ├── data/
│   │   ├── catalog.py       Step 1 – immutable Parquet vintages over DuckDB
│   │   ├── connectors/      the vintage store; `current` advances only if QC passes
│   │   ├── qc.py splice.py  fred, bis, french, jst, shiller, treasury_hqm
│   │   ├── derive.py        qc.py splice.py derive.py desmooth.py episode.py reports.py refresh.py
│   │   └── cli.py            → `ah data ...`
│   ├── gen/
│   │   ├── registry.py      Step 2 – the generator layer (research, NOT on the play path)
│   │   ├── bootstrap.py     generator_id → factory → resolve()
│   │   ├── climate/ regimes/ `bootstrap-v1`, the sealed benchmark
│   │   ├── blocks/         L1 (numpyro) and L2 (semi-Markov)
│   │   ├── joinery/        L3 – diffusion + flow (`hier-flow-v1`, torch)
│   │   └── systems.py       L4 – waypoints, bridge, Denton reconciliation
│   │       the five ablation systems A-E
│   ├── eval/
│   │   ├── prereg.py g2.py g3seal.py g5seal.py the sealed judging code (hashed in pre-registration.lock)
│   │   ├── battery.py metrics/ seals, locks, amendment log
│   │   ├── decision_metrics.py the Step-2 validation battery
│   │   └── walkforward.py counterfactual.py episode2022.py negative_controls.py Step-5 sealed metric set
│   └── port/
│       ├── bundle.py        Step 3 – the translation layer and the real institution
│       ├── feed.py
│       ├── serve.py
│       ├── play.py
│       ├── credibility.py
│       └── programme.py
```

			sleeves.py vehicles.py cohort.py	state objects
			cashflow_tier0.py cashflow_tier1.py	the commitment/call/distribution model
			portfolio.py	cash account + coverage on both bases
			engine.py	the quarterly waterfall → PortfolioEngine
			twin.py	the DB-pension institutional twin
			mapping.py smoothing.py heroes.py proxy.py	
			artifacts/	Step 4 – the wire, the letters, the committee
			templates.py render.py	Tier-1, pure functions of the tape
			author.py prompts.py gate.py	Tier-2 authoring + the G1-G9 gate
			bible.py validation.py committee.py	continuity, actor study, AI committee
			chronicle.py payloads.py	the G9 publication record (unused – §5.13)
			live.py calendar.py windows.py decisions.py	
			battery/	Step-0 stylized battery (thresholds still `todo` – §5)
			presets/	the four preset worlds
			splits.py	train/val/holdout + the FinalEvaluationToken guard (sealed)
			factors.py strategies.py	the factor manifest and the D4 strategy set (both sealed)
			experiment.py	local-first experiment tracking under experiments/
			play.py	the playable institution on the Step-3 twin → simulate_play
			bundle.py	the world bundle, contract `world-bundle-0.4` → build_bundle
			feed.py	the in-timeline tier-1 wire → build_tier1_feed
			serve.py	FastAPI session service – the scoring authority
			credibility.py	admin console: does this world's arithmetic look sane
			programme.py	the private-programme section of that console
			inspect.py	static figure page from a RunRecord alone
			tournament.py density.py	Step-5 harnesses
			cli.py exp_cli.py	→ `ah`, `ah exp`
			app/	the React/TS player (vite 5173, proxies /sessions → 8787)
			schemas/	READ-ONLY vendored contract truth
			mappings/	frozen parameter artifacts (pacing, tier-0 spec, tier-1 linkage, smoothing kernel, sleeve mappings)
			fixtures/	compiler, state, worlds, entity_screen, actor_validation, authoring_regression
			tests/	121 test modules + one shared helper
			scripts/	53 deterministic build/measure/train scripts
			Instructions/	the authoritative plans, design notes and kickoffs
			governance/	seals, amendment log, decision register, GenAI pack, evidence
			docs/	the realism register, data specs, and the superpowers plans
			artifacts/	committed research outputs (wp24...wp57, campaign2)
			configs/ taxonomy/	search configs and the sleeve taxonomy
			experiments/ (gitignored)	trained checkpoints and experiment records
			data/ (gitignored)	the Parquet vintage store and data/ah.db

Entry points, exhaustively. The console script `ah` (`pyproject.toml:35` → `ah.cli:app`); the ASGI app `ah.serve:app` (`src/ah/serve.py:318`); the module entry `python -m ah.battery.report` ; the vite app in `app/` ; and the 53 scripts in `scripts/` , each run with `uv run python scripts/<name>.py` .

2. Capability inventory

Status vocabulary: **working** — runs end to end and I ran it; **partial** — runs, but a named part of what the design asks for is absent; **stub** — present as an interface or placeholder without the substance behind it; **absent** — named somewhere but not built; **unverified** — I could not run it. Qualifiers after a status ("working, sealed"; "working locally, not reproducible from a clean clone") are free text, not vocabulary.

Flags are not exhaustive. The "how to invoke" column shows the shortest command that does the thing, not every option. Notably absent from it: `ah data refresh --dry-run/--source/--vintage` , the `--data-root / --root` overrides on the data and experiment commands, and `ah credibility --seed` . Run `--help` on any subcommand for the full set.

Coverage of this inventory, stated: `core/` , `port/` and `store/` are covered module by module. `eval/` , `gen/` , `compiler/` and `data/` are covered by capability, not by module — 28 of the 157 modules under `src/ah/` have no row of

their own, including sealed judged code (`factors.py` , `strategies.py` , `eval/reference.py` , `eval/panel.py` , `eval/ablation.py` , `eval/sleeveetails.py`) and `gen/base.py` . See §5.15.

World generation

capability	status	evidence	how to invoke
Preset worlds (4)	working	<code>src/ah/presets/*.json</code> ; ran all four through build/validate	<code>ah world build --preset stagflation</code>
Scenario → world, offline	working	<code>src/ah/compiler/fixture_adapter.py</code> ; 50 fixtures in <code>fixtures/compiler/</code> ; I built <code>00000000-0000-4000-8000-000000000000</code> from a fixture scenario	<code>ah world build --scenario "<tex from fixtures/compiler/_manifest.js</code>
Scenario → world, live LLM	unverified	<code>src/ah/compiler/anthropic_adapter.py</code> , imported lazily at <code>src/ah/cli.py:117</code>	<code>ah world build --scenario "..."</code> . live (needs network + key; not run)
Toy engine toy-v0.3	working	<code>src/ah/core/engine.py:42</code> , <code>run_path</code> at :261 , <code>run_ensemble</code> at :415	<code>ah run [WORLD_ID] --paths N --S</code>
L1 climate layer	working locally, not reproducible from a clean clone	<code>src/ah/gen/climate/</code> ; the fitted posterior is <code>experiments/climate-l1-f7d4119c7101-s20260726/climate-posterior.npz</code> , SHA-pinned at <code>joinery/assemble.py:140,146</code> . the <i>L1 fit report</i> is <code>experiments/climate-l1-.../climate-fit-report.md</code> , gitignored alongside the posterior; <code>artifacts/wp27/</code> carries the downstream joinery/battery report	<code>uv run python scripts/fit_climate.py</code>
L2 regime layer	working locally, not reproducible from a clean clone	<code>src/ah/gen/regimes/</code> ; posterior at <code>experiments/regimes-l2-1758709d4009-s20260727/</code> , SHA-pinned at <code>joinery/assemble.py:143,147</code>	<code>uv run python scripts/fit_regimes.py</code>
L3 block generators	working locally, not reproducible from a clean clone	<code>src/ah/gen/blocks/flow.py:583</code> registers <code>hier-flow-v1</code> ; the factory hard-requires <code>experiments/campaign2-flow-s0/checkpoint.pt</code> against pinned SHA <code>c6addb54...</code> (<code>flow.py:503,514,551</code>). <code>experiments/</code> is gitignored	<code>registry.resolve("hier-flow-v1"</code> — I confirmed it resolves here and the checkpoint hash matches
L4 joinery	working (research)	<code>src/ah/gen/joinery/{waypoints,bridge,reconcile,assemble}.py</code>	via the systems in <code>src/ah/gen/systems.py</code>
Ablation systems A–E	partial	<code>src/ah/gen/systems.py:503,610</code> ; registry lists <code>abl-a/b/c</code> variants. <code>abl-b-neural-rollout-diffusion</code> and <code>abl-c-neural-only-diffusion</code> are constructible but untested by design (<code>systems.py</code> §"B and C run the FLOW arm only")	<code>uv run python scripts/run_ablation_grid.py</code>
Neural generators on the play path	absent	<code>ah run</code> calls <code>run_ensemble</code> (<code>cli.py:195</code>), which calls <code>_require_toy</code> (<code>engine.py:151</code>) and raises <code>UnsupportedGeneratorError</code> for anything but <code>toy-v0</code>	— see §5.1

Compiler and coherence checks

capability	status	evidence	how to invoke
JSON-Schema validation	working	<code>src/ah/core/loader.py</code> ; schema is <code>schemas/worldspec-v1.2.schema.json</code>	inside <code>process()</code>
Pydantic mirror agreement	working	<code>src/ah/core/worldspec.py</code> ; property test <code>tests/test_worldspec.py</code>	<code>pytest tests/test_worldspec.py</code>

V1–V12 coherence rules	working	src/ah/core/validator.py:70 ; bounds clamps :128 , blocking V10/V11 :428 , V12 :493	ah world validate [WORLD_ID] — output: clamps=0 warnings=[] blocking=[]
Clamp-then-construct pipeline	working	src/ah/compiler/pipeline.py:28 ; rejection path cli.py:124	ah world build
Validation stamping	working	stamp_validation at cli.py:129 ; flips status to validated	automatic on build

Ensemble generation, replay, verification

capability	status	evidence	how to invoke
Ensemble with seed lineage	working	engine.py:427 — seeds are base_seed + 7919*k	ah run --paths 1000
Output digest	working	src/ah/core/digest.py ; compute_outputs_digest at store/runrecords.py:31	recorded on every RunRecord
Replay (bit-identical)	working	cli.py:239 ; I ran it and got MATCH on sha256:73326f13...	ah replay [RUN_ID]
Verify from lineage	working	store/runrecords.py:83	ah verify [RUN_ID] → True
Append-only chronicle	working	trigger-level refusal, store/db.py:48 ; I read a 2-entry chronicle back	ah chronicle [WORLD_ID]
World immutability	working	store/worlds.py:58 raises ImmutableWorldError on engine-consumed edits	—

Cashflow engine and the portfolio engine

capability	status	evidence	how to invoke
Tier-0 cashflow (constant-G TA)	working	src/ah/port/cashflow_tier0.py ; frozen spec mappings/cashflow-tier0-v1.0.yaml	run_tier0(...)
Tier-1 cashflow (market-linked)	working	src/ah/port/cashflow_tier1.py:90 ; linkage frozen in mappings/cashflow-tier1-v1.0.yaml ; f_dist (:48) / f_call (:61) consume continuous states only	run_tier1(...)
"Tier 1 with linkage+fees off IS tier 0"	working	asserted by test, tests/test_cashflow_tier1.py	pytest tests/test_cashflow_tier1.py
Cohort recursion	working	src/ah/port/cohort.py ; parameters in mappings/pacing-parameters-v1.0.yaml , drift-guarded by tests/test_pacing_artifact.py	—
Call curve rc_curve	partial — placeholder	[0.25,0.30,0.20,0.12,0.08,0.05] in fixtures/state/closed-end-cohort.example.json ; ~ 29% of every commitment is never called (docs/engine-realism-register.md §ER-6, status open)	— see §5.3

Quarterly waterfall	working	<code>src/ah/port/engine.py:77</code> — receive, pay calls, spend off the trailing reported average, then forced sale	<code>PortfolioEngine.run_quarter(...)</code>
Forced sale, liquid pro-rata	working	<code>port/engine.py:107-126</code> , every event logged with cause and sleeves	—
Forced secondary at haircut	working	<code>port/engine.py:128-153</code> at <code>secondary_haircut = 0.19</code>	—
Private-weight breach detection	working	<code>port/engine.py:155-170</code> , both bases	<code>QuarterReport.breach_true/breach_reported</code>
Vehicle mechanics (notice, gates, side pockets)	working	<code>src/ah/port/vehicles.py</code> ; <code>tests/test_port_vehicles.py</code>	—
Hero funds (reconciliation invariant)	working	<code>src/ah/port/heroes.py</code> ; <code>tests/test_port_proxy_heroes.py</code>	—

Decisions, twins, decision alpha

capability	status	evidence	how to invoke
Decision windows	working	<code>core/institution.py:71</code> — month <code>12y-1</code> , years <code>1-9</code> . For a 40-quarter world: <code>[11,23,35,47,59,71,83,95,107]</code> (I read this back off a live session)	<code>GET /sessions/{sid} → decision_windows</code>
Four actions	working	<code>hold / derisk / leanin / secondary</code> (<code>core/institution.py:52</code>)	<code>POST /sessions/{sid}/decisions</code>
Play institution (the scored one)	working	<code>src/ah/play.py:310</code> <code>simulate_play</code> — real cash account, a commitment ladder, calls that must be funded, forced sales	via the session service
Toy institution (Step o)	working, but still on live surfaces	<code>core/institution.py:98</code> ; used by <code>bundle.summary</code> , <code>inspect.py</code> , <code>feed.py:175</code> , <code>tournament.py</code> , <code>density.py</code>	— see §5.2
Hold-course twin (product)	working	<code>play.py:451</code> — the twin takes no action at any window	<code>simulate_play(paths, None)</code>
Hold-course twin (toy)	working	<code>core/institution.py:162</code>	<code>hold_course_twin(...)</code>
Institutional twin (DB pension)	working, v1-simple	<code>src/ah/port/twin.py</code> — parameterized benefit profile, flat discount, hedges, collateral. Full member-level projection explicitly deferred	<code>tests/test_port_twin.py</code>
Drift twin	stub — slot only	<code>serve.py:300</code> returns <code>"drift_twin": None</code> ; the contract carries three series so its arrival is a data change (E7, <code>Instructions/experience-deltas-register.md</code>)	—
Decision alpha, product	working	<code>serve.py:256</code> <code>alpha = active.final_value - twin.final_value</code> , stamped <code>PLAY_ALPHA_VERSION = "port-v1-cashflow"</code> (<code>play.py:80</code>). I measured <code>+3.3735</code> on a played decade	<code>GET /sessions/{sid}/outcome</code>

Decision alpha, research	working, sealed	eval/decision_metrics.py:55 (log points), DECISION_ALPHA_VERSION = "1.0" inside the G5 lock	pytest tests/test_decision_metrics.py
Chain-link window attribution	working	play.py:476 , K+1 exact runs, no sampling. I verified the telescoping identity live: contributions summed to 3.3735227993676062 , exactly the reported alpha	GET /sessions/{sid}/outcome → window_contributions
Counterfactual "was it a good call"	working	src/ah/eval/counterfactual.py ; tests/test_recone.py	research only
Tournament (3 formats)	working	src/ah/tournament.py — solo / cohort-cadence / simultaneous; score invariance across formats is a pinned test	pytest tests/test_tournament.py
Decision-density study	working	src/ah/density.py — but on the toy institution	pytest tests/test_density.py
Leaderboard	working (server), broken in the dev app	serve.py:212 returns rows keyed on the required triple; curl against a live service returned { "world_id": "x", "seed": 1, "decision_alpha_version": "1.0", "rows": [] }	— see §5.4

The battery

capability	status	evidence	how to invoke
Step-0 stylized battery	partial — no ratified thresholds	src/ah/battery/thresholds.yaml : every metric is status: todo , and the file says so ("Step 0 ships plumbing only... placeholders documenting intent, not ratified thresholds"). passed is not enforce_failures (battery/report.py:62), so with zero enforce entries it cannot fail	ah battery [RUN_ID] / python -m ah.battery.report — not run here, per instruction
Step-2 validation battery	working, sealed	src/ah/eval/battery.py with suites in src/ah/eval/metrics/ ; MC error bars, reference bands, prereg thresholds. Hashed into pre-registration.lock	run_full_battery(...) ; results committed under artifacts/wp27* / , artifacts/wp29/
Negative controls	working	src/ah/eval/negative_controls.py ; tests/test_negative_controls.py	—
Leakage guard	working	src/ah/splits.py ; FinalEvaluationToken mintable only in eval/g2.py ; import-graph test tests/test_leakage_guard.py . The holdout has been spent (WP5.6)	pytest tests/test_leakage_guard.py
Pre-registration seal	working	pre-registration.lock (34 hashed files), pre-registration-g3.lock , pre-registration-g5.lock ; tests/test_seal_guards.py	pytest tests/test_seal_guards.py

Artifact generation

capability	status	evidence	how to invoke
Tier-1 templates	working	src/ah/artifacts/templates.py — pure functions of the tape, no LLM/RNG/clock	via build_tier1_feed
Tier-1 wire in the bundle	working	src/ah/feed.py:159 . My stagflation bundle carried 222 items across cb_statement , newspaper , quarterly_statement , release_page , wire_digest	ah bundle [RUN_ID]

Watermarking	working	applied in the renderer, src/ah/artifacts/render.py	—
Tier-2 authoring pipeline	working, not shipped in bundles	src/ah/artifacts/author.py (two retries then Tier-1 fallback, MAX_RETRIES = 2); the ≥95% first-pass bar is met — governance/evidence/AUTHORING-REGRESSION.md records 30/30, 100.0%, "ships: YES". No bundle contains a tier-2 letter	uv run python scripts/run_authoring_regression.py (live model; not run here)
G1–G9 consistency gate	working	src/ah/artifacts/gate.py , GATE_IMPL_VERSION = "gate-impl/1.0.6" ; blocking split frozen under AM-2026-08-02-002. G3's leak-checker prompt is v1.1, not built (stated in the module docstring); G6 is advisory	pytest tests/test_authoring_regression.py
World bible	working	src/ah/artifacts/bible.py , checks B1–B6	pytest tests/test_bible.py
AI committee	working	src/ah/artifacts/committee.py ; benchmarked in governance/evidence/ACTOR-VALIDATION.md	pytest tests/test_committee.py
Live mode (sealed reveal)	working	src/ah/artifacts/live.py ; tape seal reused by the bundle (bundle.py:151)	pytest tests/test_live_mode.py

The product surface

capability	status	evidence	how to invoke
World bundle	working	src/ah/bundle.py:56 BUNDLE_VERSION = "world-bundle-0.4" ; 1 MB budget enforced at :190 . I built one: 31,263 bytes , digest_verified: true	ah bundle [RUN_ID] --out world.bundle.gz
Session service	working	src/ah/serve.py ; 7 routes. I created a session, advanced, decided at all 9 windows, completed, and read the outcome	uv run uvicorn ah.serve:app --port 8787
Mark-to-market at the pointer	working	serve.py:119 — server-side, never client-side (W5). At month 12 I read value=98.0151 , cash=0.3183 , coverage_true=0.1782 , private_weight_true=0.3436	GET /sessions/{sid}
Auth	absent by design (vo.1)	serve.py:8-10 — "no auth — single-user, local"; joins at the M4 boundary	—
React player	working	app/ ; 8 test files / 32 tests pass, tsc -b --noEmit clean	cd app && npm run dev
Client-side seal re-verification	working	app/src/lib/bundle.ts , pinned against the committed app/fixtures/toy.bundle.gz by the TypeScript suite only — the Python suite builds its own bundles and never opens that file (§5.8)	cd app && npm run test
Offline replay cache	working	app/src/lib/idb.ts (IndexedDB, W8)	—
Static figure page	working	src/ah/inspect.py ; regenerates and re-verifies the digest on every render. I produced a 125,911-byte self-contained page	ah inspect [RUN_ID] --out page.html
Credibility console (admin)	working	src/ah/credibility.py:294 ; read-only. I ran it over two presets: "2 worlds, 2 flags"	ah credibility --preset stagflation --preset goldilocks --out credibility.html
Private-programme console section	working	src/ah/programme.py:751 ; read-only enforced by import-graph test tests/test_programme_guard.py	included in ah credibility

ah audit (world register + wire audit)	absent — plan only	docs/superpowers/plans/2026-08-05-world-and-wire-audit.md (2,027 lines, 7 tasks) and its design note; ah --help lists no audit command	—
Scenario build console (WP-B, added 2026-08-06 — postdates this document's survey)	working	src/ah/buildconsole.py ; dry-run five-stage compile ledger, keep-only writes (guard test test_only_keep_handler_writes_to_store); live path known to end in validator rejection until the compiler prompt rewrite (WP-A)	uv run uvicorn ah.buildconsole:app -- port 8798
Generator-input data console (added 2026-08-07 — postdates this document's survey)	working	src/ah/dataconsole.py ; read-only over the vintage store (guard test test_dataconsole_is_read_only); raw→factor lineage, proxy shading, de-smoothing overlay, split-shaded factor panel	uv run uvicorn ah.dataconsole:app --port 8796

The data layer

capability	status	evidence	how to invoke
Requirements register	working	requirements.yaml — the single source of truth for required series	—
Immutable vintage store	working	src/ah/data/catalog.py ; re-writing a (vintage, series) raises; current advances only on QC pass	—
Connectors	working (parse tested, fetch untested by design)	src/ah/data/connectors/ — fred, bis, french, jst, shiller, treasury_hqm	ah data refresh --live (network; not run here)
Offline refresh	working	src/ah/data/refresh.py	ah data refresh --fixtures <dir> --asof YYYY-MM-DD
Status report	working	I ran it: current vintage 2026-08-02.4 , six sources, QC 138 enforce-passed	ah data status
As-of resolution	working	src/ah/data/catalog.py pointer history	ah data asof YYYY-MM-DD
Episode packs	working	I ran the 2022 pack: 42 series, window 2022-01-01→2023-12-31	ah data episode 2022
Manual intake validation	working	src/ah/data/intake.py + src/ah/data/schemas/	ah data intake validate <file> --schema albourne_pm_returns
Manual intake apply (state-mutating)	working	src/ah/data/cli.py — "Validate a drop AND apply it: write the vintage, run QC, advance on pass"; on QC failure the vintage is quarantined and the pointer stays put	ah data intake apply <file> -- vintage 2026-08-01.1 (--vintage required; --schema optional, inferred)
Experiment tracking	working	src/ah/experiment.py ; I listed 8 experiments with config hash, seed, git SHA, vintage	ah exp list / show / diff

3. The numbers as configured

Every value below is read from the file cited. **UNSET** means the code has no default and nothing in the repo supplies one.

Run shape

quantity	value	source
Horizon (all 4 presets)	40 quarters = 120 months, starting 2027-Q1	src/ah/presets/*.json
Decision windows per run	9 — months 11, 23, 35, 47, 59, 71, 83, 95, 107	core/institution.py:71 (12y-1 , years 1–9)
Ensemble size, presets	1000 paths	src/ah/presets/*.json engine_defaults.n_paths
Ensemble size, schema default	10000 (min 100, max 100000)	schemas/worldspec-v1.2.schema.json
Ensemble size, credibility console	400 paths (--paths)	cli.py:300
Seed lineage	base_seed + 7919*k	engine.py:44,427
Preset base seeds	stagflation 771204 · goldilocks 42 · deflation_bust 1848 · reflation_boom 2021	src/ah/presets/*.json
Engine version stamp	toy-v0.3	engine.py:42
Validator version	1.0.0	validator.py:27
Battery version	battery-0.1	battery/report.py:34
Bundle contract	world-bundle-0.4 , ≤1,000,000 bytes gz	bundle.py:56,57
Product alpha version	port-v1-cashflow	play.py:80
Research alpha version	1.0 (sealed)	eval/decision_metrics.py:21

Sleeves

set	members	source
Engine assets (order is contract)	equity, bonds, hy, commodities, reits, pe, pc, re	engine.py:74
Appraisal-smoothed ("reported")	pe, pc, re	engine.py:84
Liquid (play)	equity, bonds, hy, commodities, reits	play.py:82
Private (play)	pe, pc, re	play.py:83
Growth / defensive pairs	(equity, pe) / (bonds, pc)	play.py:125-126

Opening book — the scored institution

item	value	source
Targets, in points of 100	equity 33 · bonds 12 · hy 5 · commodities 5 · reits 8 · pe 20 · pc 8 · re 7	play.py:96-105
Opening cash	2.0	play.py:106
Opening private weight	35 points — deliberately inside the policy band	play.py \$START_TARGETS note

The Step-0 toy institution uses a **different** opening mix — 45 points private (core/institution.py:39), outside the play policy band. Both are live; see §5.2.

Costs, bands, policy

item	value	source
Spending rate	4.5%/yr, taken quarterly	port/engine.py:37
Spending basis	trailing 12 quarters of REPORTED value	port/engine.py:38,98

Secondary haircut (play/port)	0.19 (sells at 0.81 of NAV — the 2022-H2 public anchor)	port/engine.py:39
Secondary discount (toy institution)	0.18	core/institution.py:56
Private-weight policy band	(0.15, 0.40)	port/engine.py:40
Derisk/lean-in shift	10 points, liquid leg only	play.py:122
Voluntary secondary size	8 points of the largest live PE cohort	play.py:123
Transaction costs	UNSET — there is no cost term anywhere; LiquidSleeve.sell / buy move value at par	port/sleeves.py:203,211
Management fee / carry on the play path	supplied but never applied — every scored cohort is built from a document carrying mgmt_fee_rate 0.02, carry_rate 0.20, hurdle 0.08, European waterfall. They are never charged: cohort.step books CohortStep(..., 0.0, 0.0) for fees and carry, and the fee machinery lives in run_tier1, which the play path never calls	fixture fixtures/state/closed-end-cohort.example.json ; loaded at play.py:205-227,230-257 ; zeroed at port/cohort.py:203

Pacing

item	value	source
New vintage cadence	every 4th quarter (annual)	play.py:119
Annual commitment rate	18% of each private sleeve's target NAV (≈ 6.3 points/yr against ≈ 35 private)	play.py:118
Contractual life	10 years	mappings/pacing-parameters-v1.0.yaml
Distribution bow	2.5	mappings/pacing-parameters-v1.0.yaml
Terminal yield rate	0.55	mappings/pacing-parameters-v1.0.yaml
Call curve rc_curve	[0.25, 0.30, 0.20, 0.12, 0.08, 0.05] of remaining unfunded, per year, last value repeated	mappings/pacing-parameters-v1.0.yaml ; class "chosen", pending ALB-A
Tier-1 f_call	clip(1 - c.drawdown, 0.5, 1.2) — near-flat by design	cashflow_tier1.py:61-66
Tier-1 f_dist	clip(exp(-a.dd - b.ln(spread_ratio)), floor, ceiling)	cashflow_tier1.py:48-58 ; coefficients in mappings/cashflow-tier1-v1.0.yaml
Spread reference	400 bps	play.py:109, engine.py:60

Declared plausibility bands (admin console — priors, not truth)

Asset bands, annualized % over ten years (credibility.py:66): equity -8...14 ret / 12...28 vol · bonds -2...9 / 3...12 · hy 0...11 / 6...18 · commodities -6...16 / 12...28 · reits -8...12 / 12...26 (and pe/pc/re below those lines in the same table).

Programme bands (programme.py:209), all seven: peak_unfunded_ratio 0.25–0.75 · call_rate_y1_3 0.15–0.45 · crossover_years 4–8 · dpi_age9 0.7–2.0 · linkage_bite 0.50–1.20 · linkage_shortfall 0.05–0.35 · forced_secondaries 0.0–1.0.

Battery thresholds

All status: todo — `excess_kurtosis` , `skewness` , `hill_tail_index` , `acf_r_lag1` , `acf_abs_lag1` , `max_drawdown_median` , `corr_distance` (`src/ah/battery/thresholds.yaml`). No metric is `enforce` ; the Step-0 battery therefore cannot fail. The Step-2 battery's thresholds are real and live in `pre-registration.yaml` under the seal.

4. Test estate

Suites

suite	what it is	how to run
Python, <code>tests/</code>	121 <code>test_*.py</code> modules, 2,226 tests (plus <code>joinery_common.py</code> , a shared helper) covering core, store, data, gen, eval, port, artifacts, product surface	<code>uv run pytest</code>
TypeScript, <code>app/src/**</code>	8 test files, 32 tests (bundle loader + seal, session client, decision window, ranked setup, reckoning, fan chart, book, private markets)	<code>cd app && npm run test</code>
Type checking	pyright basic mode over <code>src + tests</code>	<code>uv run pyright</code>
Lint + format	ruff 0.16.0 (pinned in <code>uv.lock</code>)	<code>uv run ruff check . /</code> <code>ruff format --check .</code>

Structural invariants are tests, not conventions: `narrative-blindness` (`tests/test_narrative_blindness.py` scans the engine for narrative field access), `no-network` (`pytest-socket` , `--disable-socket` in `addopts`), `leakage guard` (`tests/test_leakage_guard.py` is an import-graph assertion), `seal integrity` (`tests/test_seal_guards.py`), the artifacts boundary (`tests/test_artifacts_service.py::TestBoundary` — `core / gen / port` never import `artifacts`), and the programme console's read-only guarantee (`tests/test_programme_guard.py`).

What CI runs

`.github/workflows/ci.yaml` , on push to `main` and on every PR, in order:

- `uv sync --frozen --dev`
- `uv run ruff check .` **and** `uv run ruff format --check .`
- `uv run pyright`
- `uv run pytest --cov=ah.core --cov-report=term-missing --cov-fail-under=90`
- `uv run python -m ah.battery.report`
- `uv run pytest tests/test_g0_end_to_end.py -v`

Note the coverage gate covers **ah.core only** (`pyproject.toml:69`), not the rest of the tree. Two further workflows are scheduled, not gating: `data-monthly.yaml` (a monthly public-source refresh that opens an issue on QC failure) and `data-reminders.yaml` (quarterly manual-intake reminders).

Last-known status, measured here on 2026-08-05

check	result
<code>uv run pyright</code>	0 errors, 0 warnings, 0 informations — clean
<code>uv run ruff check .</code>	All checks passed
<code>uv run ruff format --check .</code>	FAILS — 2 files would be reformatted (both markdown plan docs; see §5.5)
<code>cd app && npm run typecheck</code>	clean
<code>cd app && npm run test</code>	8 files, 32 tests, all pass
<code>uv run pytest</code>	2,226 tests across 121 modules — all passed (exit 0, ~20 minutes wall clock)
<code>uv run pytest</code> on the core rails only (engine, validator, worldspec, stores, digest, Go end-to-end)	80 passed in 3.29s
<code>python -m ah.battery.report</code>	not run — battery execution is gated pending threshold ratification (§3)

5. Known gaps and discrepancies

Ordered by how much they would surprise someone who read the design documents and then read the code. Everything here was found in this pass; nothing is copied forward from a status table without re-checking.

5.1 The promoted generator cannot reach a player

Step 2's gate promoted `hier-flow-v1` (`G2-EVIDENCE.md`; `S2-DEFAULT-GENERATOR`), and the `WorldSpec` enum carries it (`core/worldspec.py:62`). But every playable path runs the **toy engine**: `ah run` calls `run_ensemble` (`cli.py:195`), which calls `_require_toy` (`engine.py:151`) and raises `UnsupportedGeneratorError` for any other id. `ah/gen/registry.py` is never imported by `ah/cli.py`, `ah/bundle.py`, `ah/serve.py`, or `ah/play.py`. The generator layer is a research artifact; the product runs on `toy-v0.3`.

Two consequences worth stating separately:

- **A fresh clone cannot instantiate the generator stack at all.**

`hier_flow_v1_factory` requires `experiments/campaign2-flow-s0/checkpoint.pt` and refuses anything whose SHA-256 is not `c6addb54...` (`flow.py:551-557`); it then loads the L1 and L2 posteriors from `experiments/climate-11-...` and `experiments/regimes-12-...`, each SHA-pinned (`joinery/assemble.py:140-147`). `experiments/` is gitignored. It resolves on this machine; it would not on a clean checkout. Refitting/retraining is the documented alternative (`scripts/fit_climate.py`, `scripts/fit_regimes.py`, `scripts/train_flow_final.py`) and is GPU-hours of work.

- **signature-mmd is a name with nothing behind it.** It is in the schema enum

and the pydantic mirror, and the registry has no factory for it — by design (`worldspec.py:56`: "raises `UnknownGeneratorError` until something registers under it"), but a world declaring it is constructible and unrunnable.

5.2 Two institutions are live at once, and both reach the player

`ah/play.py` (cash account, calls, forced sales) is what scores. `ah/core/institution.py` (eight weights, no cash, nothing can be owed) is Step 0's toy. The toy is still wired into surfaces the player sees:

- `bundle.py:98` computes the **toy** twin (`hold_course_twin`) and `:160` writes

its final value into `summary.twin_final_value` (and `run` puts the same number into `summary_stats.final_of_100`, `cli.py:224`);

- `bundle.py:158` puts the **port** twin's ledger into `twin_ledger`;
- `feed.py:175,180` builds the quarterly institution statement and its peer bands

from `simulate_institution` — the toy;

- `inspect.py`, `tournament.py` and `density.py` all run the toy.

This is not theoretical. In the single bundle I built from the stagflation preset (run `03a76c08...`, seed 771204, 200 paths):

```
summary.twin_final_value      = 166.502   (toy institution)
twin_ledger.nav_reported[-1]  =  76.712   (port institution – the one that scores)
```

The same file states the same institution's terminal value twice, 2.2× apart, and the wire the player reads is narrated off the higher one. The bundle docstring's own framing ("the HOLD-COURSE TWIN's cashflows") is accurate for `twin_ledger`; `summary.twin_final_value` is the leftover.

5.3 ER-6: about 29% of every commitment is never called

`docs/engine-realism-register.md` §ER-6, status **open**, dated 2026-08-05. The call curve is an ALB-A placeholder that was never fitted; measured at `committed = 1.0`, paid-in reaches 0.7075 over the contractual life against an allocator's 85–95% expectation. Downstream, `peak_unfunded_ratio` flags on all four presets (3.018 / 2.445 / 3.308 / 2.453 against a declared 0.25–0.75) and the J-curve crossover lands at ~8.5 years against a declared 4–8. On `deflation_bust` no year-1 vintage crosses over within the decade at all. The register is explicit that this is a prerequisite for the commitment lever (E1), not a parallel cleanup.

Two further register entries are open: **ER-2** (the policy rate is a continuous drift with no meeting calendar and no 25bp quantisation) and **ER-5** (the crisis is a rectangular block, giving equity an ACF of 0.364). ER-1 and ER-4 are closed in `toy-v0.3`; **ER-3 is closed in the play surface** (`wire-play-surface`) — a different layer, since ER-3 was the no-cashflow gap and was fixed in `ah/port/`, not in the return process.

5.4 The leaderboard is unreachable from the dev app

`app/src/lib/session.ts:176` fetches `/leaderboard/{worldId}`. `app/vite.config.ts:8` proxies **only** `/sessions`. Verified against the running dev stack:

```
GET http://127.0.0.1:8787/leaderboard/x?seed=1&alpha_version=1.0 → 200 application/json
GET http://localhost:5173/leaderboard/x?seed=1&alpha_version=1.0 → 200 text/html (vite's SPA fallback)
GET http://localhost:5173/sessions/nope → 404 application/json (proxied correctly)
```

The client parses the SPA's HTML as JSON and throws; `Leaderboard.tsx:31` catches it and renders the error string in place of the board. The server side is fine — the same request direct to 8787 returns a well-formed empty board. It is a one-line proxy entry.

5.5 `ruff format --check .` fails on `main` today

`ruff 0.16` formats Python code blocks inside Markdown, and two committed plan documents contain unformatted blocks:

- `docs/superpowers/plans/2026-08-04-private-programme-console-section.md`
- `docs/superpowers/plans/2026-08-05-world-and-wire-audit.md`

`uv.lock` pins `ruff 0.16.0` and CI runs `uv sync --frozen`, so CI's lint step would take the same view. `ruff check .` and `pyright` are both clean; this is the formatter alone, and `ruff format` on those two files fixes it.

5.6 The battery CI step cannot fail

`.github/workflows/ci.yml` runs `python -m ah.battery.report` as a gate, but every threshold in `src/ah/battery/thresholds.yaml` is `status: todo`, and `BatteryReport.passed` is not `enforce_failures` (`battery/report.py:58-64`). With no `enforce` metric the step is green by construction. The file is candid about this ("placeholders documenting intent, not ratified thresholds") — the gap is that CI presents it as a gate.

The CI step is also mislabelled. It is titled "Validation battery (stagflation preset)", but `main()` runs `_stagflation_ensemble()` (`battery/report.py:189-196`), which loads `schemas/example-long-stagflation.worldspec.json` — a schema example, not `src/ah/presets/stagflation.json` — forces `generator_id = "toy-v0"`, and runs **64 paths at `base_seed=42`**, not the preset's 1000 at 771204.

Note the Step-2 battery (`ah/eval/battery.py`) *does* have real, sealed thresholds and all eight suites registered (`_REFERENCE_DEPENDENT_SUITE_BUILDERS`, `battery.py:955`); the two are different objects with similar names.

5.7 Tier-2 authoring passes its bar and still ships nothing

`governance/evidence/AUTHORING-REGRESSION.md` records the current run as **30/30, 100.0% first-pass, "ships: YES"** against the frozen $\geq 95\%$ bar, and `governance/evidence/G4-EVIDENCE.md` closes criterion 1 at 96.7%/100.0%. But every bundle still stamps:

```
meta.artifact_tier = "tier-1 templated wire (build-time, deterministic);
                    tier-2 letters await the frozen >=95% first-pass bar"
```

(bundle.py:139-142 , and I read it back out of a freshly built bundle).

The string is stale, but the deeper point is that it describes the wrong rule. The ratified decision, PD-4

(Instructions/KICKOFF-PRODUCT-SU.md:78), is **key-conditional, not bar-conditional**: "tier-1 deterministic always; tier-2 letters *when a key is present at build*, recorded in the bundle either way." bundle.py has no such branch — it never imports ah.artifacts.author , so a key present at build time would change nothing. The stale string is the symptom; the missing PD-4 branch is the gap.

Also unbuilt: G3's leak-checker prompt over the draft, recorded as v1.1 in artifacts/gate.py 's docstring, and G6 which warns "not evaluated" unless past artifacts are supplied.

5.8 Documentation that no longer matches the code

claim	where	what the code says
bundle contract is world-bundle-0.3	CLAUDE.md \$Architecture	bundle.py:56 — world-bundle-0.4
" ah/pacing.py is a display-only toy ledger"	CLAUDE.md \$Architecture	the module was deleted in 168cd22 ("retire ah/pacing.py"); only a stale .pyc remains. The pacing table lives in mappings/pacing-parameters-v1.0.yaml
docs/tier1-synthesis-and-decisions.md , named by Step 2's vendoring list	CLAUDE.md	still missing — confirmed absent
"CLI-echoed strings stay ASCII"	CLAUDE.md \$Environment gotchas	data/episode.py:107 emits an em-dash, echoed via typer.echo(pack.brief) at data/cli.py:154 . It does not crash — exit 0 under PowerShell with the dash rendering correctly — but it renders as ? in Git Bash. (data/reports.py 's em-dashes are not a deviation: they are in the module docstring and in the generated GAPS.md , which CLAUDE.md explicitly permits Unicode in)
" app/fixtures/toy.bundle.gz is a committed bundle both suites verify"	CLAUDE.md \$Regenerating fixtures, repeated in CHANGELOG.md	only the TypeScript suite reads it (app/src/lib/bundle.test.ts:22). No file under tests/ mentions it; tests/test_bundle.py builds its own bundle. The two seal implementations are not pinned against a shared fixture — which is the property the sentence is claiming

5.9 Gate state, carried forward and re-checked

These are the project's own recorded positions, verified against the evidence files rather than restated from the status table:

- **Step 3's G1-completion is an honest FAIL** — G1-EVIDENCE.md : mark_lag

(must-pass) failed at 0.9667 and private_weight_breach failed at 0.0332. Tier 1 beat tier 0 (2 criteria failed vs 3). **G3 itself was never taken.**

- **The holdout is spent** — declined at G2 on purpose, spent at WP5.6.

RESEARCH-EVIDENCE.md : primary drawdown surprise −0.3952, realized terminal wealth at the 99.6th percentile of the ensemble, cpi band coverage 0.000, and fx_usd went unread because of a reader fallback bug that, per the sealed spec, publishes as a gap rather than re-running.

- **The severe test is inconclusive** and the L3 leg structurally vacuous

(0 blocks dropped) — artifacts/wp211/SEVERE-TEST.md via RESEARCH-EVIDENCE.md .

- **The standing caveat** — `hier-flow-v1` beats the benchmark on the sealed

criterion and is not a convincing model of history: regime persistence undercalled, drawdowns understated $\sim 2\times$, the decade tier 73% structurally unavailable (`G2-EVIDENCE.md` §7–8). Nothing built on it is decision-ready.

5.10 Named-but-absent, and deferred-by-decision

- **ah audit** — 2,287 lines of design and implementation plan merged at

`545a2ac` ; no code, no CLI command.

- **Drift twin** — the third analysis series is a `None` in the API contract

(`serve.py:300`), by design, so its arrival is a data change.

- **E1 commitment lever** — `Instructions/experience-deltas-register.md` marks

E1, E2 and E4 **PARTIAL**, E3 and E5 **OPEN**, E6 **OPEN** (I5's ~ 20 -user first-run observation deferred under D-K4-5), E7 and E8 **CLOSED**.

- **Auth** — deliberately absent at v0.1 (`serve.py:8-10`); required before any

external user.

- **Human-cohort benchmark and the too-rational pathology** — deferred by owner

decision D-K4-5, recorded in `G4-EVIDENCE.md` criterion 4/5 rather than proxied with a number.

- **Fee/carry on the play path** — `simulate_play` steps cohorts directly

(`play.py:385`) and never calls `run_tier1` , so the management fee, the European carry, recycling, subscription-line deferral and extension behaviour that `cashflow_tier1.py` implements are **not applied to the scored institution**. Whether that is intended is not stated anywhere I found.

- **Preset structural coverage** — three of four presets carry only

`parameter_vintage` under `structural` ; the toy engine falls back to `_DEF` (`engine.py:89-106`) for private-market parameters in those worlds.

5.11 A secondary sale relieves nothing — the obligation survives it

This is on the scored path, and it inverts the sign of the only liquidity lever the player has.

Both the voluntary sale (`play.py:298-307`) and the forced secondary (`port/engine.py:136-140`) reduce `cohort.nav_true` and `cohort.nav_reported` and credit cash. **Neither touches unfunded or committed**. Nothing else does: in

`port/cohort.py` , `unfunded` moves only in `step()` (:200) and `recall()` (:233); `report()` (:207-211) writes the mark and returns. And calls are sized `call = min(self.unfunded, call_rate * self.unfunded)` (`cohort.py:185`) — a function of the unfunded balance, independent of NAV.

So you sell an LP interest at a 19% haircut, keep 100% of the future call obligation, and — since coverage is `unfunded ÷ what you own` — **coverage gets strictly worse after the one action that exists to relieve liquidity stress**. A cohort driven toward `nav_true ≈ 0` by repeated forced secondaries keeps calling at full size. Not in `docs/engine-realism-register.md` (ER-1...ER-6 do not reach it).

5.12 Two writers to the leaderboard, and a version-label collision

The `decision_alpha_version` column is the mechanism that is supposed to keep incompatible scores off one board (`db.py:92` , `UNIQUE(world_id, seed, decision_alpha_version, participant)`). Three numeric definitions exist and two of them share a label:

writer	label	institution	metric
serve.py:257	port-v1-cashflow	port (cash account)	terminal value difference
tournament.py:268	1.0 (from store/runrecords.py:27)	toy (§5.2)	terminal value difference
eval/decision_metrics.py:21	1.0 (sealed under G5)	—	log points, $\ln(\text{player}/\text{twin})$

tests/test_play.py:167 asserts only that `PLAY_ALPHA_VERSION` differs from the sealed constant; nothing pins `runrecords.py`'s constant against it. The comment at `runrecords.py:22-24` ("inert... Nothing reads them yet") is stale — `tournament.py` reads it and writes it to the board. Every bundle also stamps `meta.decision_stamps.decision_alpha_version = "1.0"`.

5.13 Named-but-unwired, beyond §5.10

- `src/ah/artifacts/chronicle.py` — 88 lines implementing artifact

publication as `type='artifact'` `chronicle` entries, described as the G9 record. Its only importer anywhere is `tests/test_artifacts_service.py`. §5.7 lists G3 and G6 as the unbuilt gate pieces; G9's record is unwired too.

- `scripts/build_artifact.py` executes

`Path("scratch_data.json").read_text(...)` **at module import** (:9). That file is not in the tree and never was (`git log -all` finds nothing). The script cannot be imported, let alone run — yet `CLAUDE.md` still lists it under "Regenerating committed fixtures/artifacts (all deterministic)".

- `ah data intake apply` — "Validate a drop AND apply it: write the vintage,

run QC, advance on pass." A state-mutating command with no row in §2, which documents `intake validate` only.

5.14 Pacing is open-loop, and one fixture governs all three private sleeves

Three separate simplifications, none of them registered:

- **The commitment size never responds to anything.**

`amount = START_TARGETS[asset] * _ANNUAL_COMMITMENT_RATE` (`play.py:213`) is a constant. Not a function of current NAV, the world, the coverage ratio, or any decision. A portfolio halved by a crash keeps committing ~6.3 points a year, driving private weight up mechanically at the worst moment.

- **PC and RE run buyout's cashflow profile.** `play.py:334` loads one base

document and stamps `sleeve_id` onto copies (`play.py:217`); `mappings/pacing-parameters-v1.0.yaml` carries exactly one sleeve row, `pm_buyout`. Private credit and real estate therefore inherit buyout's `rc_curve`, bow 2.5, terminal yield 0.55 and 10-year life. There is no distinct PC or RE cashflow shape in the model at all.

- **Equity drawdown drives every sleeve's linkage.** `play.py:337-339` computes

`drawdown_depth` from `q_returns["equity"]` alone and feeds the resulting `f_call / f_dist` to `pe`, `pc` and `re`.

ER-6 covers the *shape* of `rc_curve`. None of these three is in the register.

5.15 Smaller things that still change what a reader concludes

- **The decision actions are absolute point amounts that can silently truncate.**

`_SHIFT_POINTS = 10.0` and `_SECONDARY_POINTS = 8.0` are value units on a book that starts at 100 and ends anywhere; at the ninth window on a book worth 76, "10 points" is 13% of it. Worse, `_rebalance` returns silently when the source leg is empty (`play.py:272-277`) and the secondary caps at available NAV (:301) — so a chosen action can execute partially or not at all and surface as a `0.0` window contribution **indistinguishable from hold**.

- **CI never runs the app, and never runs on Windows.** `ci.yml` has no Node

step: the React suite (32 tests), its typecheck, its build, and the client-side seal implementation are all ungated. And `runs-on: ubuntu-latest` only — so the ASCII-CLI rule in §5.8 could not be enforced by CI even in principle.

- **CI never validates the promoted checkpoint.** Six tests skip when

`experiments/` is absent — `tests/test_blocks_flow.py:689` ("no primary flow checkpoint present (pre-training checkout)"), `:806`, `:832`, `:855`, `test_blocks_diffusion.py:481`, `test_blocks_tuning.py:204`. On a clean CI clone all six skip. The "2,226 passed" figure in §4 is this machine's; CI sees fewer, and none of them exercise `hier-flow-v1`'s checkpoint.

- **28 of 157 modules under `src/ah/` have no inventory row**, including sealed

judged code: `factors.py` and `strategies.py` (both in `pre-registration.lock`), `eval/reference.py` (2,540 lines, sealed in both `pre-registration.lock` and `pre-registration-g3.lock`), `eval/panel.py`, `eval/ablation.py`, `eval/sleevetails.py`, and `gen/base.py`. Worth noting against §5.9's "G3 itself was never taken": G3-pre's judged code exists and is sealed.

- **`README.md` still describes the repository as Step o.** "This repository

contains **Step o** — 'lay the rails'... *No real market data, no ML training, no UI, and no LLM in the numeric path.*" Three of those four negations are now false (there is a real vintage store, there are trained models, there is an app); only "no LLM in the numeric path" still holds, and it is a standing invariant. Its architecture tree lists six components and omits `data/`, `gen/`, `eval/`, `port/`, `artifacts/` and `app/`. It is the repository's front door.

- **`app/README.md` is stale** — it describes `App.tsx` as "the v0.1 shell

(browse mode; session binding is `su-app-02`)" and lists four files, written before `su-app-01...05` merged.

- **The governance registers are not read by this document.**

`governance/retrofit-register.md` is an append-only log of **96 dated deferrals**, and neither this summary nor the manual cites it, the decision register, the amendment log, the model inventory, the prompt registry, or the EU AI Act mapping. One row bears directly on §5.9's standing caveat: **RFR-66** records that the G2 head-to-head is biased *toward* promotion by the benchmark's data window — `bootstrap-v1` can only resample 1990–2020 while the challenger saw 1929–33, 1937, 1973–74 and 1987, and both are scored against realizations that include all of it. It was sealed into the pre-registration rather than deferred. Anyone auditing §5 for completeness should read that file next; it is the project's own gap register and it is longer than this section.

Evidence for every "working" claim in §2 is either a file:line in this repository or a command I ran on 2026-08-05; the commands and their real output are in `docs/USER-MANUAL.md`. Anything I could not verify is marked, and the full list of what I could not verify is in the closing note of that document.